A Distributor-Centric Blockchain Framework for Regulatory-Aware Pharmaceutical Supply Chain Verification

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***Abstract*—*The pharmaceutical supply chain has problems because fake medicines, old medicines, medicines that have been taken back and medicines that are not allowed by rules end up with patients. Current ways to check medicines mostly use computer systems and old ways of keeping records. These ways can only show an information about where the medicine came from the information is not all together and it takes too long to follow the rules. This paper suggests a system called a Distributor-Centric, Regulatory-Aware Blockchain Enforcement System for Pharmaceutical Verification. This system helps track medicines and makes sure rules are followed all through the process from the distributor to the pharmacy to the patient. Every medicine has a QR code. This code helps check if the medicine and the batch are real if they have expired, who owns them if they are allowed by rules and what happened with them in the supply chain. A special database keeps track of what's happening with medicines and batches. Blockchain keeps a record that cannot be changed of events like getting the medicine moving it stopping it and giving it to a patient. Smart contracts make sure rules are followed so that old not allowed, taken back not allowed or wrong medicines do not move through the supply chain. Pharmacies can check again. Patients can scan the QR code themselves to see information, about the medicine. The system suggested here gives a way to check medicines that can be followed and seen clearly. It can be tested with medicines, fake medicines, old medicines, duplicate medicines and not allowed medicines.*

Keywords*—*Blockchain, pharmaceutical supply chain, counterfeit drug detection, smart contracts, QR code verification, regulatory compliance, CDSCO, traceability.

I. Introduction

The pharmaceutical supply chain is one of the safety-critical logistics networks in existence directly affecting patient health outcomes. In India making sure that genuine, valid and regulatory-compliant medicines reach patients remains a serious challenge. Counterfeit, expired, recalled and banned drugs still appear in distribution networks. Sometimes these unsafe medicines reach pharmacies and patients before any enforcement action can be taken. These problems come from systems for tracking and enforcing pharmaceutical movement across different stakeholders. Each stakeholder often keeps its records and has little visibility into what others are doing. Today's traceability methods mostly depend on databases and manual record-keeping. These systems have a point of failure and are vulnerable to tampering. Regulatory updates move slowly because there is no connection between CDSCO notifications and inventory status. Information stays scattered among participants, including patients who have no way to independently verify if a medicine is authentic when they buy it.

Blockchain technology provides a resistant shared ledger that can improve supply-chain traceability. Many existing proposals store all data, on the blockchain and treat regulatory status as something fixed and unchanging. This approach fails to reflect how a medicine's status can change suddenly after a recall. It does not allow real-time updates when new alerts are issued.

This paper proposes a Distributor-Centric, Regulatory-Aware Blockchain Enforcement System that is built on three principles. The first principle says that verification must cover the distributor–pharmacy–patient workflow checking every custody transition. The second principle keeps decision-making separate from the blockchain storage. An updatable database tracks each batch's status—permitted, restricted, prohibited, recalled or unknown—based on CDSCO notifications and the blockchain records and enforces supply-chain events. The third principle uses contracts to automate enforcement. These contracts validate transactions against expiry dates, ownership and regulatory status. They stop the movement of any batch whose status changes after the batch has entered the chain.

Every medicine carries a QR code that lets anyone verify its status at each stage. Distributors check stock with the QR code. Pharmacies verify the code before dispensing the medicine. Patients can also scan the QR code on their own to see the medicine's status extending transparency to the link, in the chain.

II. Literature Review

Counterfeit and substandard medicines continue to be one of the serious problems that affect global healthcare. The World Health Organization reports that one out of ten medical products available in low- and middle-income countries is either not good enough or fake. This leads to 30.5 billion US dollars being wasted each year by health systems. Importantly it causes treatments to fail makes bacteria stronger and leads to deaths that could have been prevented [1], [2]. The World

Health Organizations global data also shows that most of the cases reported are about anti-malarial medicines and antibiotics. Also the way these cases are reported is not the same in all areas. There are gaps in the technology used to find these problems and in the way regulations are followed, even when a problem is found [3], [4]. Studies on diseases show the same pattern. For example research on anti-tuberculosis medicines found that weak checking after a medicine is on the market lets bad or unapproved medicines stay in use even after regulators have taken action [5]. All this information shows an idea that is the reason for this work. The problem of unsafe medicines is not only, during the making. A medicine can be real when it starts moving through the supply chain but still become dangerous or not follow rules later after a recall, a ban or a safety warning. Therefore any checking system must check for rules all the time not once.

Blockchain technology has become the technical solution to this problem over the past few years. This is mostly because it can't be changed once data is added and it runs on a network without a control. These features make it perfect for keeping a record of a medicine's journey through the supply chain. At first people used blockchain with QR codes on products. This helped track a medicine from the factory all the way to the person who buys it. If someone tried to change the record or move the medicine in a way that wasn't logged it would show up as a problem. This basic idea. Using a shared ledger and a scannable code on the product. Has been used again and again in ways. Some systems focus on logging who owns the product at each step like when it moves from the maker to the distributor or to the store. This helps stop people from changing the records without permission. Other systems go further. Track the medicine from the very start of production all the way to the pharmacy. This gives a picture of where the medicine has been and who handled it. Another group of projects uses permissioned blockchains, like Hyperledger Fabric. These networks are private. Only allow known players in the pharmaceutical industry to join. The goal here is to keep control over who can see or add data than letting anyone check it publicly. Some systems also add data from sensors into the blockchain.

For example if a vaccine needs to be kept cold sensors can check the temperature. Send that info to the blockchain. If the temperature goes too high it's. Can be spotted as a possible sign of tampering. More recent efforts have improved the QR code itself. Of just being a simple code it now has built-in encryption to make it harder to copy or fake. These protected codes are put right on the medicine packaging. At the time patients can verify the product using a simple interface. This is similar, to the role-based check system suggested in the plan. These new methods make it much harder to fake medicine or trick the system.

Across this cluster of work what becomes clear is a shared assumption. Verification is treated as validating a product against the product's recorded transaction history. When a medicine is scanned the product is checked to see whether the batch, number and ownership chain of the product are internally consistent. If they are consistent the product is considered trustworthy. Broader surveys of the field confirm that this design pattern is the prevailing one, not a choice. Several studies explicitly flag this as a limitation. A tamper-resistant ledger can prove that the product's recorded history has not been altered. It cannot on its own tell you whether the product is still legally permitted to be sold today. In words traceability and regulatory compliance are treated as separate concerns, in most existing systems even though a fully trustworthy verification result depends on both.

This gap is mentioned in reviews of blockchain applications in the broader healthcare area. Reviews that cover patient-record management, data integrity and drug traceability keep showing that most blockchain-healthcare projects are still at the prototype stage and not real systems that connect with outside data sources [6], [7]. More recent reviews of blockchain-based health information systems come to the conclusion and also point out machine-learning-based anomaly detection as a good but not yet fully explored way to improve verification past fixed rule-based checks [8]–[10]. This idea is very important for the future work of the system being proposed since detecting anomalies in serial-number use and transaction patterns is seen as a next step once the main regulatory-synchronization structure is set up.

Because the system being proposed like the ones mentioned above uses Solidity contracts to handle supply-chain tasks like batch registration and blocking the reliability of the contract layer itself is also an important design issue. Security studies of Ethereum contracts list common types of flaws. Reentrancy, integer overflow and access-control problems for example. And the techniques that check for them before they are put into use [11]–[14]. These studies are helpful more, as a list of problems to avoid when creating important contract logic because if a smart contract is broken it would destroy the trust that the rest of the system depends on. Finally on the side the Central Drugs Standard Control Organisations Drugs Rules require QR-code and barcode labelling for pharmaceutical products under Schedule H2 [15]. This regulation is important because it sets the compliance basis for using QR-based identification in the context. Yet it only specifies a labelling requirement, not a technical architecture for how that QR code should be verified against changing regulatory status once a product is in the market.

Synthesizing across these strands of literature shows a pattern instead of a bunch of unrelated gaps. The scale-of-problem studies [1]–[5] show that danger still exists after a product enters the market. The blockchain traceability systems [16]–[25] check identity and history. Do not check current legal status. The systematic reviews [6]–[10] confirm that this is a trend across the field not just a single mistake. The smart contract security literature [11]–[14] deals, with a separate layer of trust. The regulatory framework [15] sets a labelling obligation but does not say how it should be enforced in real time. The proposed system sits at the intersection of these gaps. By pairing a Regulatory Data Synchronization Module. Which keeps pulling in prohibition, recall and alert information. With a multi-factor Verification and Enforcement Engine that checks serial validity, expiry, ownership, supply-chain history and regulatory status all at once the system builds on the blockchain

traceability model that was reviewed earlier. It can flag a medicine that was legal yesterday. Has become non-compliant today closing the specific gap that shows up again and again in the literature.

III. Methodology

The suggested model uses a distributor-focused and regulation-oriented approach in blockchain technology to verify the identity of pharmaceutical products following their entry into the market. The model differs from simple traceability models in that it reassesses the regulatory and supply chain situation of the product every time it is scanned, making it possible for the product to be detected as having expired or been recalled, as illustrated in Fig. 1.

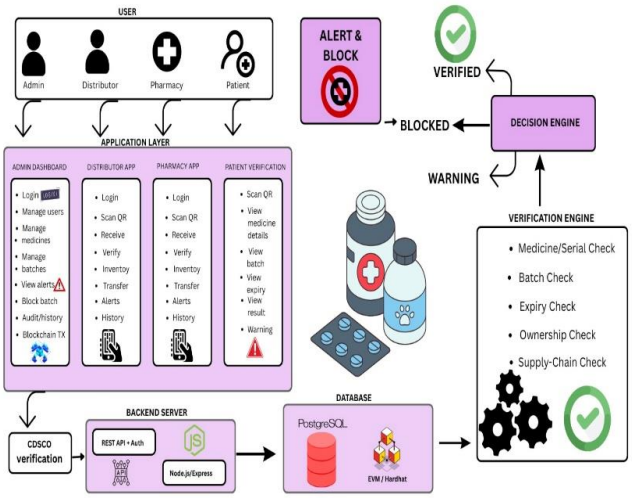


Fig. 1. Architecture diagram of the proposed distributor-centric, regulatory-aware blockchain verification system.

A. System Architecture

The system consists of four entities, namely, distributor, pharmacy, patient, and administrator, which are interconnected through a centralized backend. Each medicine will be associated with QR/barcode information (name of medicine, batch number, serial number, manufacture/expiry date). Scan will be provided to the Verification and Enforcement Engine, which will output either of these three statuses: VERIFIED, WARNING, or BLOCKED.

B. Regulatory Data Synchronization

A separate module is responsible for regulatory knowledge base that is retrieved from reliable sources (such as CDSCO) and includes prohibited/banned drugs, recall and alert information. The data goes through collecting, extracting, normalizing, and storing phases and then is stored in PostgreSQL as an approved, restricted, prohibited, recalled, or unknown.

C. Medicine Registration and QR Verification

Each medicine/batch is registered through QR or barcode. When scanned, the identifier is recognized and matches the records from PostgreSQL. It ensures the multiple verification at various phases such as receiving, transferring, dispensing, and patient level.

D. Verification and Enforcement Engine

The engine performs the following checks in sequence:

1) Medicine/Serial check – validity and duplicate serial number
2) Expiry check – expiry date comparison with the current date
3) Regulatory check – matching with the regulatory database
4) Ownership/Inventory check – authorization of the requesting organization
5) Blockchain traceability check – consistency with transaction history

The Decision function incorporates all these checks as follows in (1):

$$D = f(S, E, R, O, B) \quad (1)$$

where D is the decision and S, E, R, O and B are the outcomes of the serial, expiry, regulatory, ownership and blockchain checks. Critical failure cases such as regulatory issues or expiry result in BLOCKED, minor inconsistencies lead to WARNING while successful check results in VERIFIED.

E. Regulatory Enforcement Module

If a regulatory agency at some later point in time bans/recalls any batch currently in circulation, then the new regulatory status overrides the supply chain approval on the following scan, leading to a BLOCKED status – independent of blockchain history.

F. Blockchain Traceability Module

The blockchain traceability module consists of an Ethereum-compliant blockchain (Hardhat for local testing) together with a Solidity smart contract where events related to batch registration, batch receipt, batch transfer, batch block and dispensing are registered. Only traceability data is registered on the blockchain while sensitive data stays off-chain.

G. Alert and Enforcement Module

Alerts are generated for conditions such as prohibited/recalled products, expired medicines, duplicate serials, ownership mismatches, and supply-chain inconsistencies. Administrators can review and investigate flagged records through a centralized dashboard.

H. Role-Based Application Module

The role-based access control (RBAC) determines access rights: distributors and pharmacists take care of receiving, distributing, dispensing, and stock management; patients receive a user-friendly public verification module; regulators are responsible for the regulatory records, user management, alerts, and blockchain monitoring.

I. Security and Audit Mechanism

JWT token-based authentication, password hashing, RBAC, input validation, and logging protect the application layer. The private keys to the blockchain are separated from user

interfaces, and all verification/enforcement processes are logged.

J. End-to-End Workflow

The entire verification process is done according to the following workflow: Medicine in the Market → QR/Barcode Scanning → Product Identification → Regulatory Check → Expiration Date Check → Serial Number Check → Blockchain Check → Decision Engine → VERIFIED/WARNING/BLOCKED.

IV. Results and Analysis

The prototype of Pharma-Chain was tested on a local development server via all its user interfaces, including Login, Admin, Distributor, Pharmacy, and Patient. The screenshots given below illustrate what each screen represents for the project.

A. Access Control and Administration

The users log into the system using the login screen (Fig. 2) and get directed to dashboards relevant to their role. The administrator is presented with system-level modules, while the distributor has access to only stock management and verification modules. The administrator dashboard (Fig. 3) shows 80 medicines, 17 batches, and 34 open alerts, along with recent recalls displayed next to batch status. Batches mentioned in the alerts of recalls are blocked, hence any regulatory notice would automatically block the batch. The medicine catalog (Fig. 4) contains all types of single active ingredients medicines and fixed-dose combination medicine in one data format.

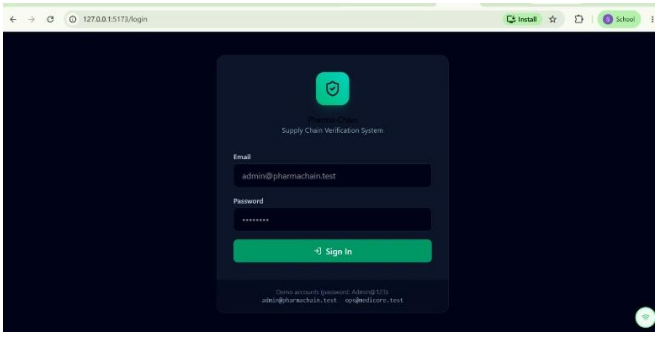


Fig. 2. Login page of Pharma-Chain.

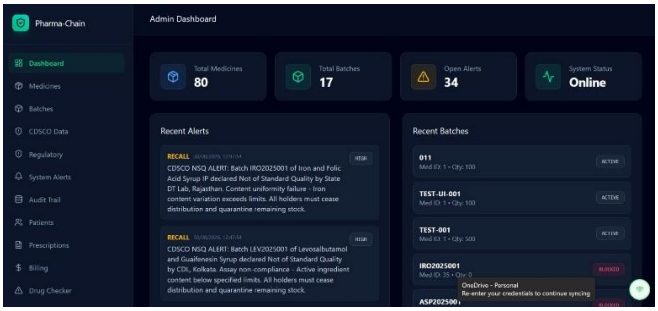


Fig. 3. Administrator dashboard with statistics, recall alerts and batch status.

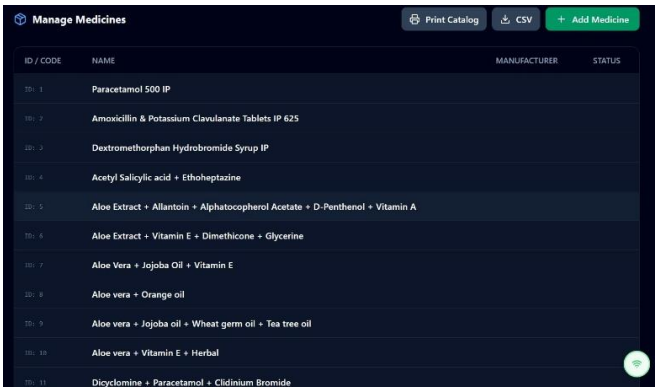


Fig. 4. Medicine management catalogue.

B. Regulatory Data Integration

The CDSCO data module (Fig. 5) includes 32 records that are regulatory actions, 22 being FDCs that are not allowed and 10 being NSQ or recalled drugs. Integration of these sources of information into the system helps to automatically update medications and batch IDs for each regulatory action.

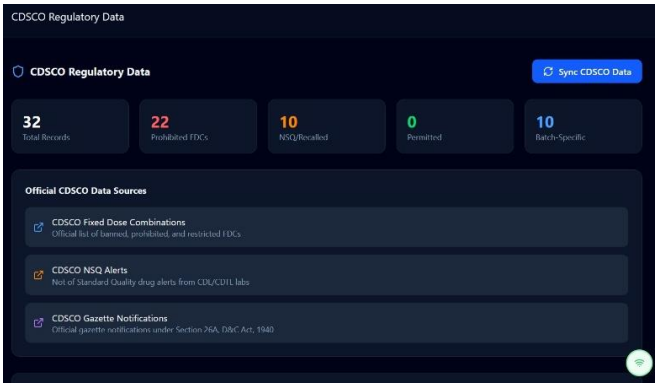


Fig. 5. CDSCO regulatory data module.

C. Distributor and Pharmacy Operations

Distributor inventory (Fig. 6) includes data on the inventory of serial-level drugs and two out of six products are BLOCKED so that recalled inventory will be seen prior to distribution. Serials of the drugs have individual QR codes (Fig. 7) that include verification link while the internal scanner (Fig. 8) scans the pack using camera or image file and compares it to the central backend system. Audit trail (Fig. 9) includes receive events in chronological order but records marked "blockchain offline" were saved only in the database. Pharmacy dashboard (Fig. 10) includes inventory levels and critical recalls notifications while the product that has been recalled has a card with regulatory notice (Fig. 11).

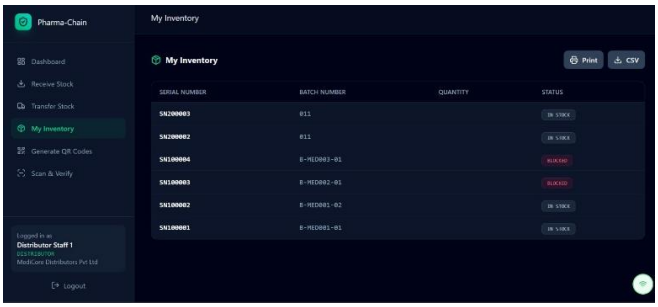


Fig. 6. Distributor inventory with stock status.

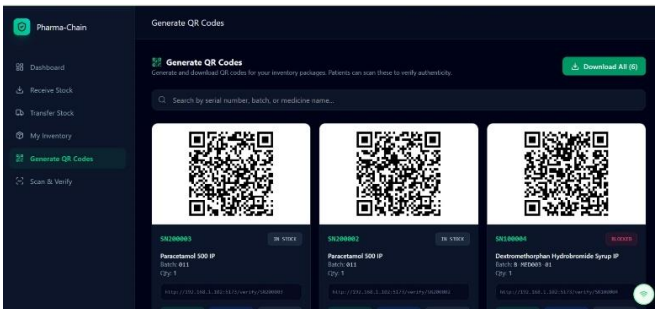


Fig. 7. QR code generation for individual packages.

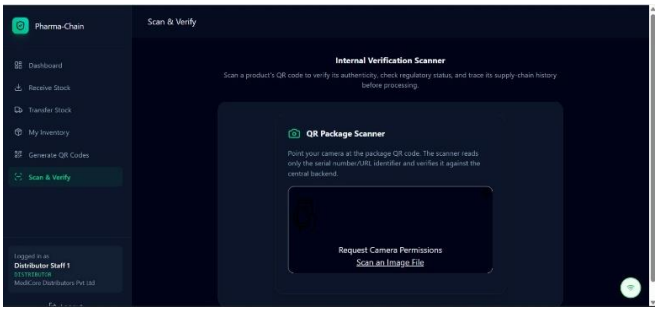


Fig. 8. Internal scan and verification interface.

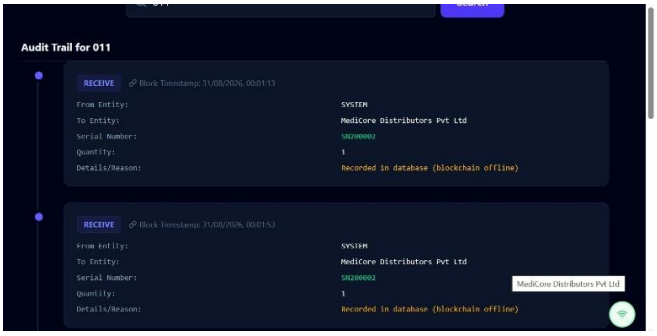


Fig. 9. Audit trail of batch movement events.

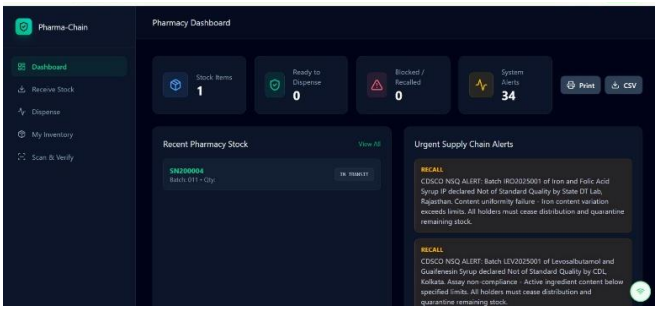


Fig. 10. Pharmacy dashboard with recall alerts.

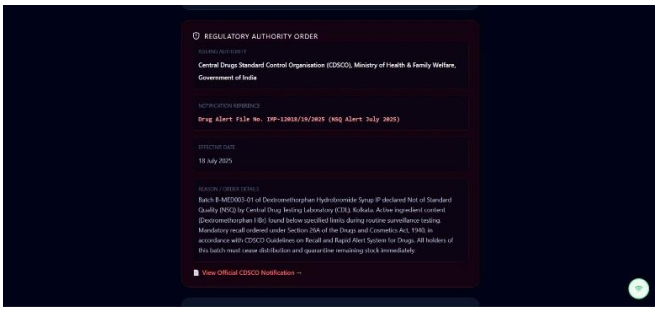


Fig. 11. Regulatory authority order for a recalled batch.

D. Results of Patient Verification

A verification test consists of eight independent tests: serial number, batch, drug, expiration date, regulation, ownership, chain of distribution and dispensing. The authentic package (SN100001) successfully passed all of them and the banner AUTHENTIC & SAFE appeared (Figs. 12 and 13). The expired package (SN100002) remained a legitimate medicine and the test result revealed only expiration date failure with a message not to dispense it (Figs. 14 and 15). A failed test reveals the exact problem due to separate test reporting. Table I presents the results.

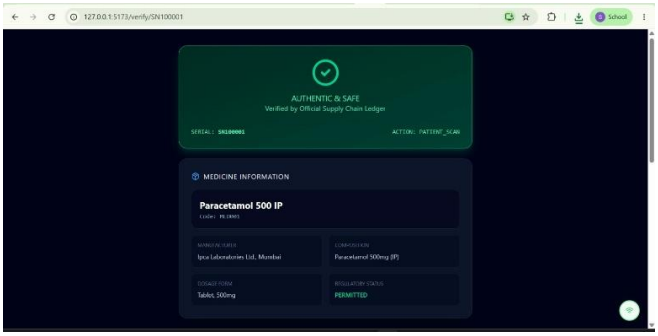


Fig. 12. Verification result for a genuine product.

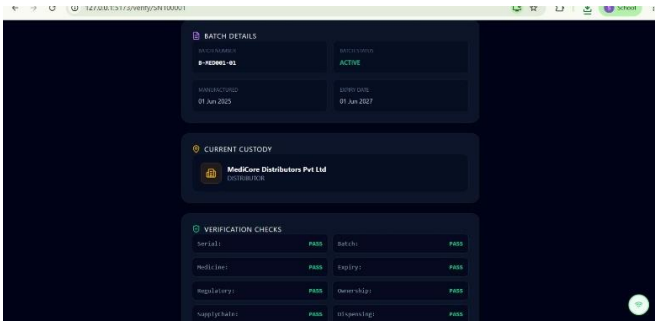


Fig. 13. Batch details and passed checks for SN100001.

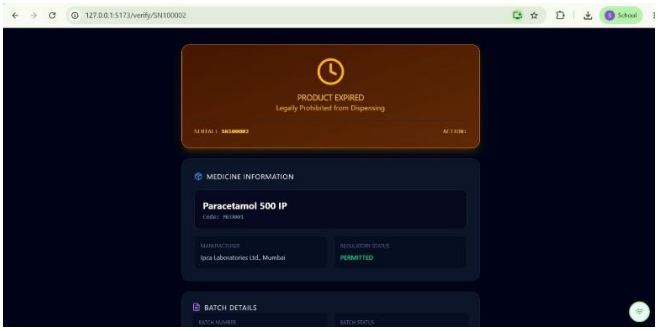


Fig. 14. Verification result for an expired product.

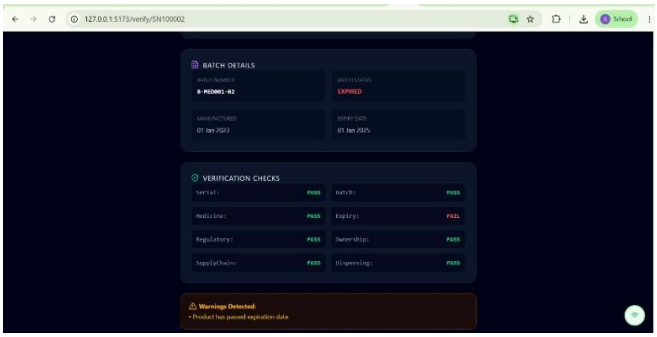


Fig. 15. Batch details with failed expiry check for SN100002.

TABLE I
SUMMARY OF VERIFICATION OUTCOMES

Serial	Batch	Failed Check	Outcome
SN100001	B-MED001-01	None	Authentic & Safe
SN100002	B-MED001-02	Expiry	Expired, do not dispense
SN100004	B-MED003-01	Regulatory	Recalled, quarantine

E. Limitations

The system integrates the regulatory information, batch sheets, and package identification codes, so that dangerous products can be marked during all stages. There are four main limitations. Firstly, the audit log records made when the blockchain was off-line must be synchronized for tamper-resistance. Secondly, the QR links contain the network IP address and demand a public HTTPS domain. Thirdly, some fields in the interface were empty, such as expired product action and some columns in the list. Lastly, the performance was tested with a demo data set only.

V. CONCLUSION

This project presents a novel use of blockchain to ensure safety and traceability of medication delivery from its manufacture to final consumption. Every item is marked using a QR code or barcode and then evaluated by a central verification engine based on information stored in the PostgreSQL database and blockchain network. Different parties have varying permissions according to their responsibilities. Each scan triggers evaluation of medicine identity, batch identification number, serial number, expiration date, owner and movement history of a particular item. In case of duplication of serial numbers, unrecognizable product, or unusual transaction, the engine marks such an entry and delivers one of three possible outcomes: VERIFIED, WARNING, or BLOCKED. Products that are considered as BLOCKED are prevented from further distribution throughout the chain.

The use of blockchain ensures the validity of transactions that should be recorded in an immutable ledger since records in blockchain are extremely hard to change. On the other hand, the database is responsible for processing bigger volumes of regular application data. During the process of prototyping, the system was able to detect multiple types of pharmaceutical violations and block them from further distribution. Taken together, these results suggest the framework can bring greater openness, traceability, and responsibility to pharmaceutical distribution. The prototype was tested only in a controlled setting, so it should be seen as a starting point. It lays the groundwork for wider real-world rollout and for smarter, more automated monitoring of medicine supply chains in the future.

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